RESULTS

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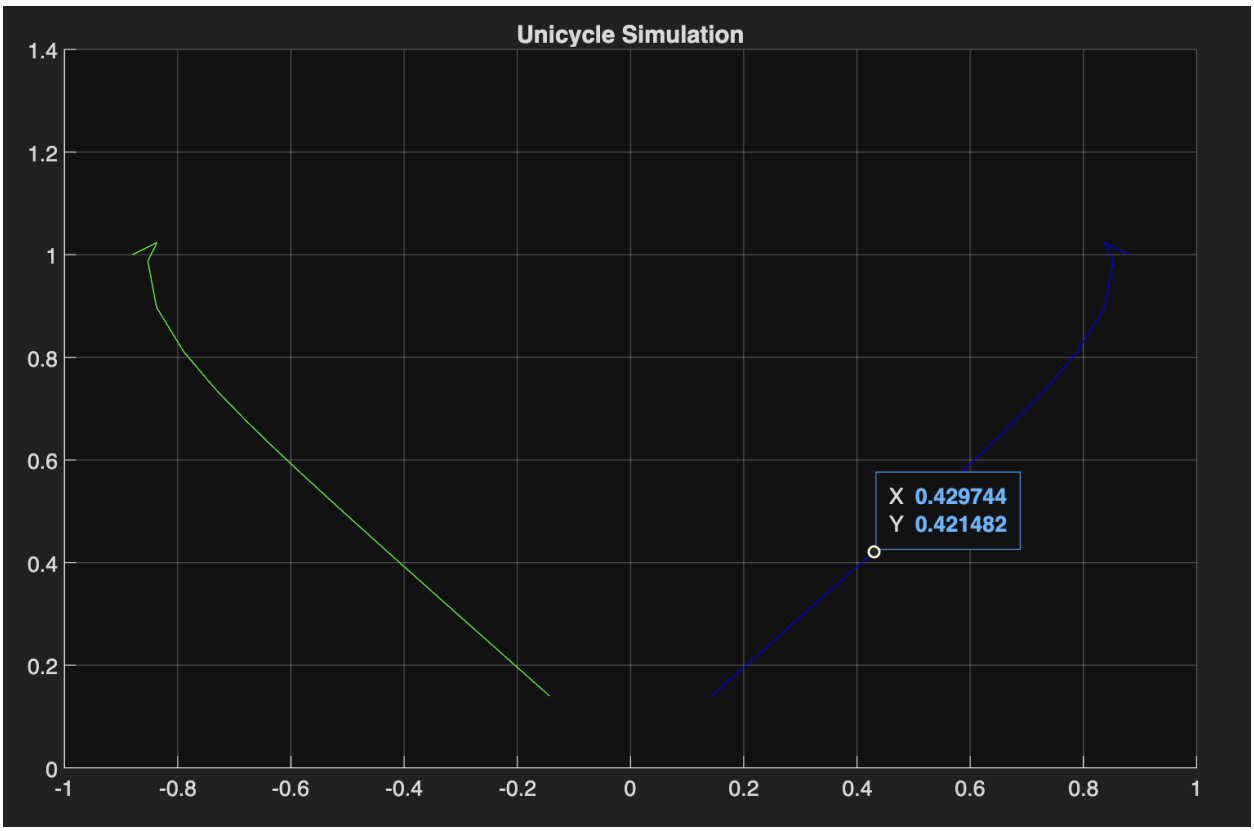
1) Unicycle Model



**This is a snapshot of the result, the actual result is a simulation.
The simulation shows the trajectory of the unicycle model under the given control law. The paths indicate convergence behavior, where the system moves smoothly toward a stable configuration from different initial directions.

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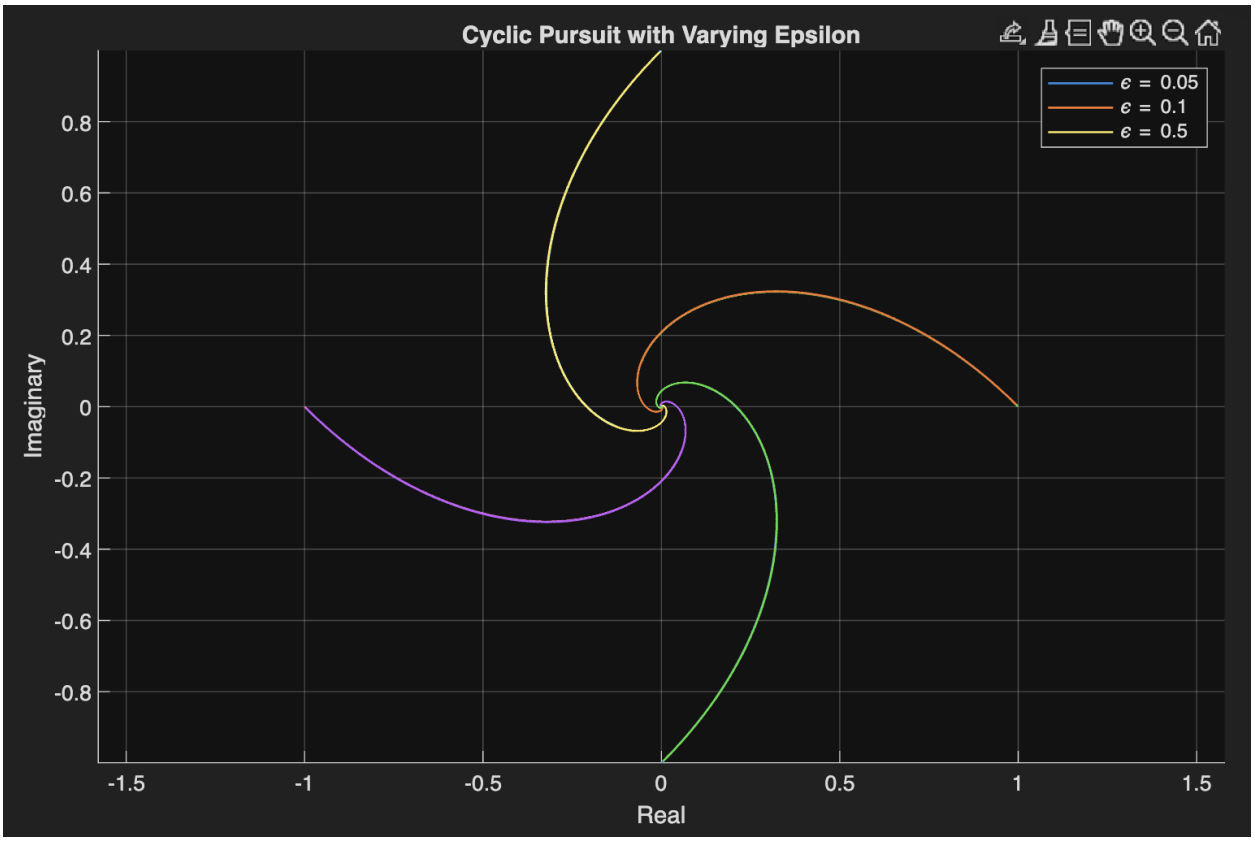
2) Cyclic Pursuit



The plot illustrates cyclic pursuit dynamics, where all agents converge to a common equilibrium. Higher ε values result in quicker convergence and reduced spiral radius, demonstrating increased system responsiveness.

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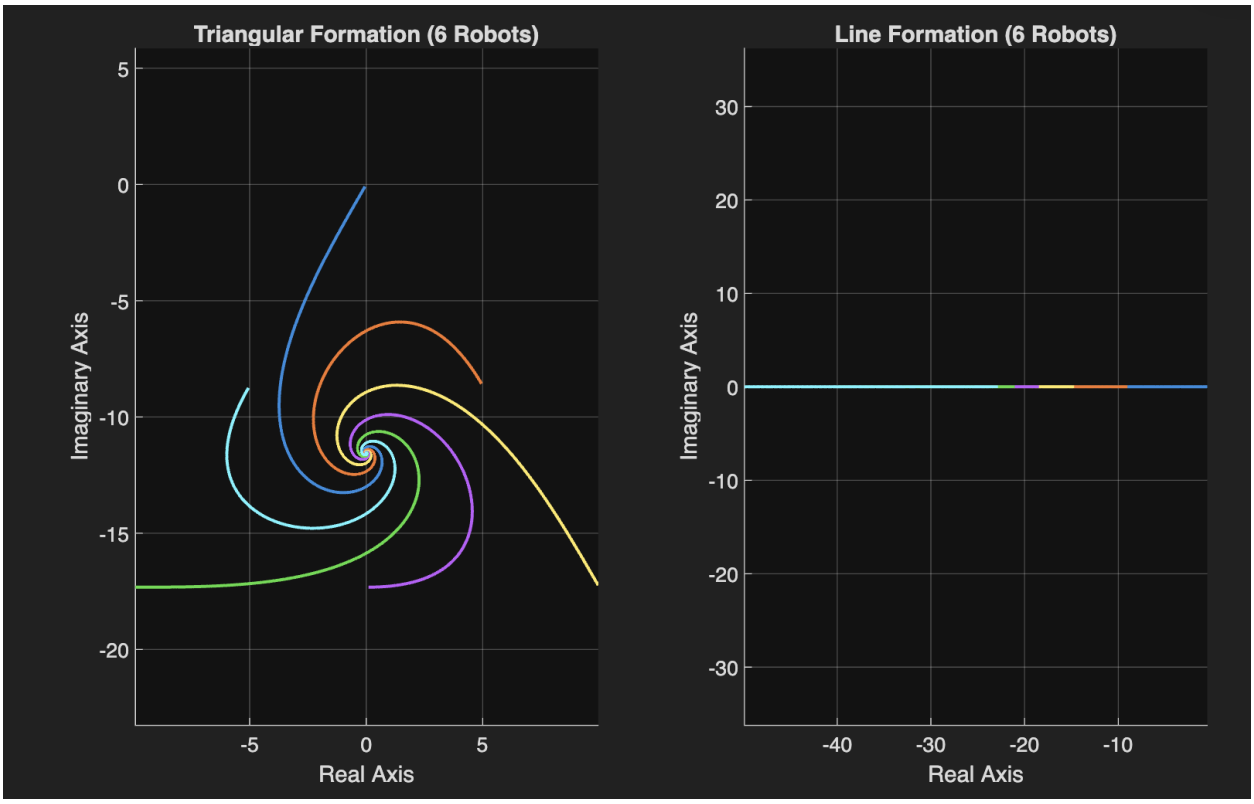
3) Triangle and Line Formation



The simulation shows that six robots under cyclic pursuit converge to the desired formations. Starting from triangular and linear initial configurations, the agents follow spiral trajectories and stabilize into the specified triangular and line formations defined by the vectors c and d (this is an example from the Flocking and Rendezvous in Distributed Robotics book).

c and d vectors are mentioned below (for triangle and line):

$$c = \begin{bmatrix} 0 \\ 5 - j5\sqrt{3} \\ 10 - j10\sqrt{3} \\ -j10\sqrt{3} \\ -10 - j10\sqrt{3} \\ -5 - j5\sqrt{3} \end{bmatrix}, \text{ which gives } d = \begin{bmatrix} -5 + j5\sqrt{3} \\ -5 + j5\sqrt{3} \\ 10 \\ 10 \\ -5 - j5\sqrt{3} \\ -5 - j5\sqrt{3} \end{bmatrix}.$$

For the line:

$$c = \begin{bmatrix} 0 \\ -10 \\ -20 \\ -30 \\ -40 \\ -50 \end{bmatrix}, \text{ which gives } d = \begin{bmatrix} 10 \\ 10 \\ 10 \\ 10 \\ 10 \\ -50 \end{bmatrix}.$$